

CMSE 801 - Day 11

NumPy Arrays

October 10, 2022

General Course Announcements

- How the Homework assignment 2 is going?
- Midterm will be on Wednesday Oct 19 during the class
- The details of the semester project have been released

Questions/comments from Day 11 PCA

- What is "cmap" when viewing the RGB portions of our jpeg? The '?' function says that cmap is not necessary for RGB(A) data, but if I take it out the result is different when I run the code. Is the type of RGB we have not RGB(A)? How can we tell what to put in the "cmap" value?
- loading images are still kind of confusing to me.
- Everything about loading images and interpreting arrays from them. I'm very confused.
- Could you introduce more of how to use colon in class? Thanks.
- Explanation from the beginning of how to load images and how to interpret arrays from them. The code supplied was incredibly confusing to me.
- If we can load picture and edit it in python, so maybe we can also apply it on other things like music?
- OMG, I just find a new method to edit the picture in python without putting it into Photoshop, especially the colors which have confused me for a long time. It is really inspiring. Thank you!
- The overall numpy material was good and it's now starting to click (also thank you for including the old videos again - great refresher!).
- I liked masking!

```
In [1]: import numpy as np
```

```
x = np.random.rand(5,5)
print(x)
```

```
[[0.0933704  0.28236363 0.61398769 0.01390977 0.88790048]
 [0.44764387 0.37328627 0.29279773 0.88456844 0.96287953]
 [0.45443228 0.94386376 0.09322697 0.59047397 0.22978702]
 [0.59017402 0.53019178 0.94380024 0.16501498 0.8189384 ]
 [0.94731399 0.62995475 0.25677343 0.39662155 0.71872855]]
```

```
In [2]: # first column
print(x[:,0])
```

```
# second row from component 2 to the end
print(x[1,1:])
# second row from component 2 to 3
print(x[1,1:3])
```

```
[0.0933704  0.44764387 0.45443228 0.59017402 0.94731399]
[0.37328627 0.29279773 0.88456844 0.96287953]
[0.37328627 0.29279773]
```

In [3]:

```
# copy
y = x.copy()
x[0] = 2.0
print(y)
print(x)
```

```
[[0.0933704  0.28236363 0.61398769 0.01390977 0.88790048]
 [0.44764387 0.37328627 0.29279773 0.88456844 0.96287953]
 [0.45443228 0.94386376 0.09322697 0.59047397 0.22978702]
 [0.59017402 0.53019178 0.94380024 0.16501498 0.8189384 ]
 [0.94731399 0.62995475 0.25677343 0.39662155 0.71872855]]
[[2.         2.         2.         2.         2.         ]
 [0.44764387 0.37328627 0.29279773 0.88456844 0.96287953]
 [0.45443228 0.94386376 0.09322697 0.59047397 0.22978702]
 [0.59017402 0.53019178 0.94380024 0.16501498 0.8189384 ]
 [0.94731399 0.62995475 0.25677343 0.39662155 0.71872855]]
```

In [4]:

```
a = 100*x[1,1:]
b = a.round()
print(b)
print(type(b[0]))
b = b.astype(int)
print(b)
print(type(b[0]))
```

```
[37. 29. 88. 96.]
<class 'numpy.float64'>
[37 29 88 96]
<class 'numpy.int64'>
```

In [5]:

```
mask = x >= 0.5
mask1 = (x>=0.5) & (x<0.75)
print(x)
print(mask)
print(mask1)
```

```
[[2.         2.         2.         2.         2.         ]
 [0.44764387 0.37328627 0.29279773 0.88456844 0.96287953]
 [0.45443228 0.94386376 0.09322697 0.59047397 0.22978702]
 [0.59017402 0.53019178 0.94380024 0.16501498 0.8189384 ]
 [0.94731399 0.62995475 0.25677343 0.39662155 0.71872855]]
[[ True  True  True  True  True]
 [False False False  True  True]
 [False  True False  True False]
 [ True  True  True False  True]
 [ True  True False False  True]]
[[False False False False False]
 [False False False False False]
 [False False False  True False]
 [ True  True False False False]
 [False  True False False  True]]
```

In [6]:

```
y = x[mask]
print(x.shape)
```

```
print(y)
print(y.shape)
```

```
(5, 5)
[2.          2.          2.          2.          2.          0.88456844
 0.96287953 0.94386376 0.59047397 0.59017402 0.53019178 0.94380024
 0.8189384  0.94731399 0.62995475 0.71872855]
(16,)
```

```
In [7]: from PIL import Image
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [8]: picture = Image.open("example.jpeg")
plt.imshow(picture)
plt.axis('off')
```

```
Out[8]: (-0.5, 499.5, 363.5, -0.5)
```



```
In [9]: print(type(picture))

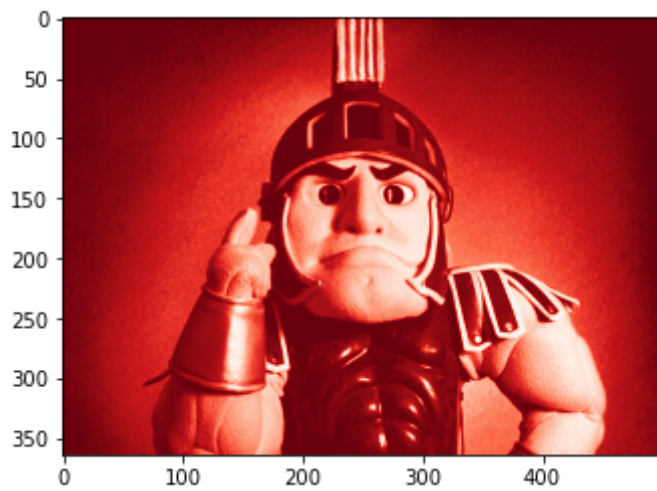
<class 'PIL.JpegImagePlugin.JpegImageFile'>
```

```
In [10]: img_array = np.asarray(picture)
print(type(img_array))
print(img_array.shape)
print(img_array[:, :, 0])
```

```
<class 'numpy.ndarray'>
(364, 500, 3)
[[ 0  0  0 ...  1  1  1]
 [ 0  0  0 ...  0  0  0]
 [ 0  0  0 ...  0  0  0]
 ...
 [ 0  0  0 ... 19 18 11]
 [ 0  0  0 ... 21 15  9]
 [ 0  0  0 ... 22 15 12]]
```

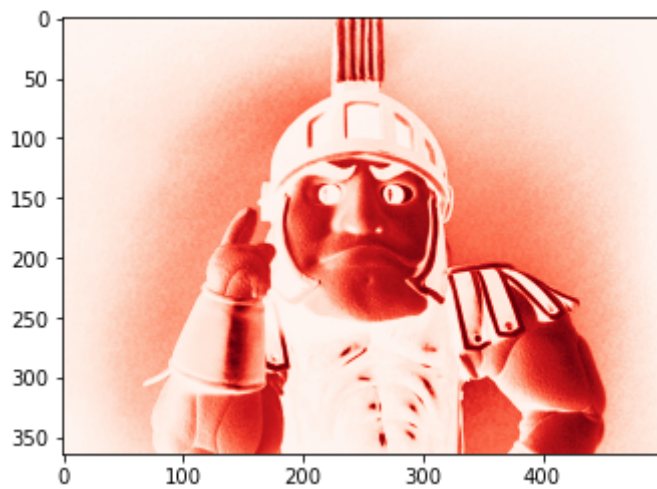
```
In [11]: red_array, green_array, blue_array = img_array[:, :, 0], img_array[:, :, 1], img_array[:, :, 2]
plt.imshow(red_array, cmap = "Reds_r")
```

```
Out[11]: <matplotlib.image.AxesImage at 0x7feb7985a550>
```



```
In [12]: plt.imshow(red_array, cmap = "Reds")
```

```
Out[12]: <matplotlib.image.AxesImage at 0x7feb79996520>
```



```
In [13]: plt.imshow(red_array, cmap = "gray")
```

```
Out[13]: <matplotlib.image.AxesImage at 0x7feb7992fd90>
```

